Earth Jukebox — Game Plan

Guardian of Dhaka · NASA Space Apps Challenge 2026 · 26 September 2026

Overview

Earth Jukebox: Guardian of Dhaka is a short survival game where players protect Dhaka from real heat waves, using NASA climate data turned into music. It is our project for the NASA Space Apps Challenge 2026.

Game goal: protect Dhaka from heat waves from 2000 to 2050.

Real goal: help people understand how cities heat up, what cools them, and how climate change makes it harder, using real NASA data.

Main rule: you can't win unless you learn. Knowledge is your weapon.

The twist: the game gets harder because Earth really got hotter. The difficulty comes from NASA data, not from us.

The player first learns how to play, then what each sound means, then plays the real game. The whole game takes about 9–10 minutes.

Part	What happens	Time
Start	Meet Echo, enter your name	30 sec
Part 1: How to play	Learn tap, drag and listen	1 min
Part 2: Learn the sounds	Play in the Sound Lab	2 min
Practice check	Quick Ears test	30 sec
Part 3: Guardian of Dhaka	6 rounds, from 2000 to 2050	5 min
End	Final future and boarding pass	30 sec

When they finish, the player should know 4 things:

1. How to tell normal from abnormal in climate data.
2. Concrete traps heat, and trees and water cool a city.
3. Cities have limited money, so choices matter.
4. Heat waves are getting stronger, but our choices can change the future.

Start (30 sec)

The game opens on a 3D Earth spinning in space, while Echo, a small satellite, beeps.

Echo says: “Hi! I'm Echo. I turn Earth into music. Heat waves are coming to Dhaka, and I can hear them. Will you help me protect the city?”

The player types their name. The name appears on the boarding pass at the end.

Part 1: How to play (1 min)

Echo teaches the 3 controls one at a time. The player must try each control once before moving on.

Control	Echo says	Player does
Drag	“Drag over the city and listen to it!”	Moves a finger over the 3D city and hears the sound change
Listen	“Hear that high note? That's danger!”	Finds the hot area by sound
Tap	“Tap to place a fix!”	Plants a tree on the hot spot

Echo's last tip: “Use headphones. Sound is the key!”

Part 2: Learn the sounds (2 min)

In the Sound Lab, Echo shows what each sound means, and the player learns by playing with them.

Sound	Means	Normal	Something is wrong
Melody	Heat	Calm, middle notes	High and sharp
Drums	Rain	Steady beat	Very fast (flood) or silent (drought)
Soft chords	Trees and green areas	Sweet sound	Out of tune

> **Golden rule:** calm music means normal. Strange music means something is wrong.

The player moves 3 sliders: heat, rain and trees. The 3D Earth changes color with each slider, so they see and hear the change together. To finish the Sound Lab, they complete 3 missions:

1. Make a heatwave (push heat up).
2. Make a flood (push rain up).
3. Make Earth calm again (bring everything back to normal).

Practice check: Quick Ears (30 sec)

Five short sounds play one after another, and the player taps heat, rain or trees for each one.

- **3 or more correct:** the game unlocks.
- **Wrong answer:** Echo plays the sound again and gives a hint.

Part 3: Guardian of Dhaka (5 min)

The player protects Dhaka through 6 rounds: 2000, 2010, 2020, 2024, 2035 and 2050. Rounds up to 2024 use real NASA observations. Rounds 2035 and 2050 use NASA future projections.

One round (about 50 seconds)

1. **Listen:** Earth's song plays, and the heat melody starts rising in one part of the city. The player hears the danger before they see it.
2. **Find:** the player drags over the 3D city and follows the sound to where the heat wave will hit.

3 **Protect:** the player has limited coins and 20 seconds to place fixes.

4 **Heat wave hits:** protected areas stay cool. Unprotected areas turn red, schools close and people get sick.

5 **Echo's report:** Echo shows the real NASA data for that year, for example: *"This really happened in April 2024."*

After each round, the city grows and more concrete appears.

Fixes the player can use

Fix	Cost	What it does
Tree	1 coin	Cools a small area
White roof	2 coins	Stops a building from trapping heat
Water (pond or lake)	3 coins	Cools a bigger area
Cooling center	4 coins	Keeps people safe, but does not cool the city

Coin costs and cooling amounts are game estimates based on NASA data. The game shows this on screen.

Getting harder every year

Each round's heat wave comes from the real data for that year. Later rounds are harder because Dhaka really got hotter, and there is less free space as concrete spreads.

Win and lose

- **Score:** the share of Dhaka the player protected, shown as a percentage.
- **Lose:** if more than half the city is unprotected in a round, the player loses that round and tries again.

Final round: choose the future

Before 2050, the player makes 2 big choices: clean energy or coal, and more parks or more concrete. These choices decide which 2050 they hear: Future A (calmer) or Future B (hotter). Both futures come from real NASA projections.

End: Boarding pass and points

The player gets a boarding pass with their name, final score (for example, *"You protected 82% of Dhaka"*), the future they reached, and their badges. They can download or share it. A QR code on the pass links back to the game, so each shared pass brings in new players.

Reward	How it works
Score	Share of Dhaka protected across all rounds
Stars	1 to 3 stars per round
Badges	Heat Hunter (find every heat wave fast), City Cooler (protect 80% or more), Future Maker (reach Future A)
Replay	Play again to beat your score

Data note

Only the data changes; the game stays the same. NASA releases the full challenge statements and datasets on October 28, and the event runs November 14–15, 2026 (source).

NASA data	The player protects the city from
Heat	Heat waves
Rain or floods	Floods
Fire or smoke	Fires and smoke
Forests	Forest loss

Until the datasets come out, build and test the game with free NASA temperature data (NASA POWER) for Dhaka.